5 Joint Probability Distributions

5.1 Joint Probability Distributions for Two Random Variables

Definition. Let $X,\ Y$ be two random variables. With a ordered pair $(X,\ Y)$, we define a **joint probability distribution function** and denote

$$f_{XY}(x,\ y).$$

In discrete case, we called **joint probability mass function**; In continuous case, we called **joint probability density function**.

Property. Let $X,\ Y$ be two random variables.

1. If $f_{XY}(x,\ y)$ is a joint probability mass function, then it satisfies the following properties:

   (1) $f_{XY}(x,\ y) \geq 0$

   (2) $\displaystyle\sum_{x\in X}\sum_{y\in Y} f_{XY}(x,\ y) = 1$

   (3) $f_{XY}(x,\ y) = P(X = x,\ Y = y) = P(X = x\ \ \&\ \ Y = y)$

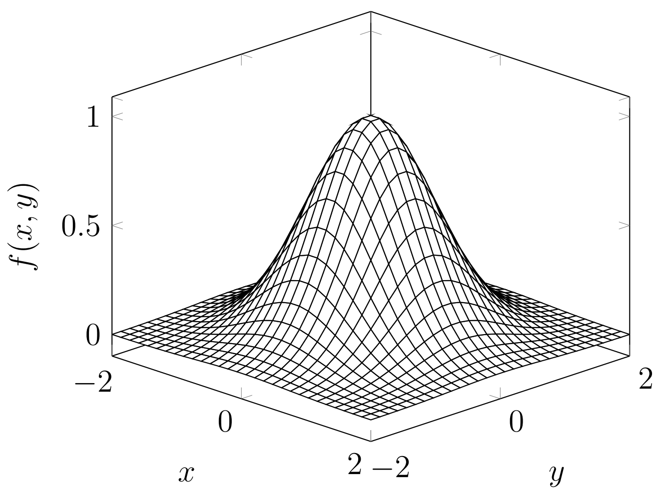
2. If $f_{XY}(x,\ y)$ is a joint probability density function, then it satisfies the following properties:

   (1) $f_{XY}(x,\ y) \geq 0$ for all $x,\ y$

   (2) $\displaystyle\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(x,\ y)\,dx\,dy = 1$

   (3) For any region R of two-dimensional space,

$$f_{XY}(x,\ y) = P((X,\ Y) \in \mathbb{R}) = \iint_R f_{XY}(x,\ y)\,dx\,dy.$$



Definition. (Marginal Density) Let $X,\ Y$ be two random variables.

1. For discrete case, the marginal(邊際) density is defined as

   (1) $f_X(x) = \sum_{y \in Y} f_{XY}(x,\ y)$

   (2) $f_Y(y) = \sum_{x \in X} f_{XY}(x,\ y)$

2. For continuous case, the marginal density is defined as

   (1) $f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,\ y)\, dy$

   (2) $f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,\ y)\, dx$

Example. Suppose $f_{XY}(x,\ y)$ is a joint probability mass function with two random variables $X = \{0,\ 1,\ 2\}$ and $Y = \{0,\ 1,\ 2,\ 3\}$. It is as shown in the table below

$Y = y$	$X = x$		
	0	1	2
0	0.15	0.1	0.05
1	0.02	0.1	0.05
2	0.02	0.03	0.2
3	0.01	0.02	0.25

Then the marginal probability distributions of X and Y are

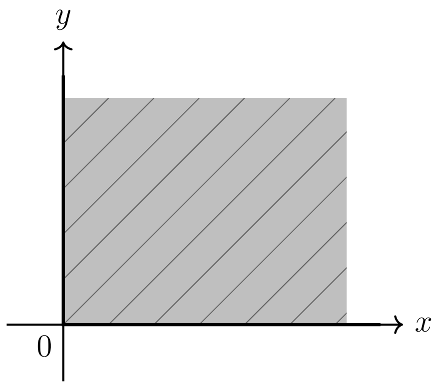
$Y = y$	$X = x$			
	0	1	2	$f_Y(y)$
0	0.15	0.1	0.05	**0.3**
1	0.02	0.1	0.05	**0.17**
2	0.02	0.03	0.2	**0.25**
3	0.01	0.02	0.25	**0.28**
$f_X(x)$	**0.2**	**0.25**	**0.55**	1

Example. Let $f_{XY}(x,\, y)$ be a joint probability density function and be defined as

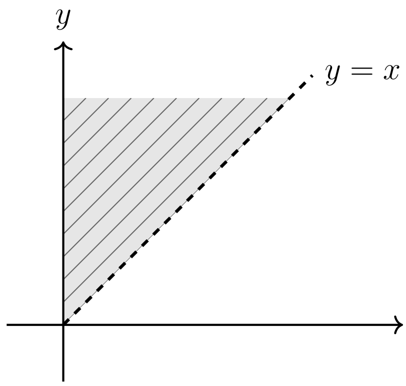
$$f_{XY}(x,\ y) = ke^{-0.01x-0.02y} \text{ for } 0 < x < y < \infty,$$

where $k = 6 \times 10^{-6}$. The region with nonzero probability is shaded in below. Find the $P(Y > 2000)$.

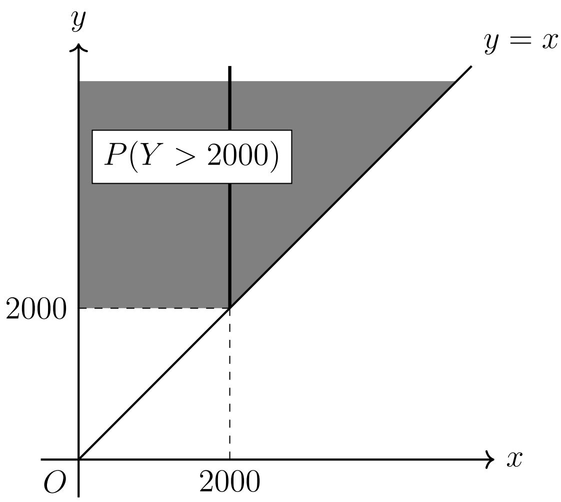
(i) **Region :** $0 < x < \infty,\ 0 < y < \infty$



(ii) **Region :** $0 < x < y < \infty$



Sol. The region is shaded in below.



So

$$
\begin{aligned}
P(Y > 2000) &= \int_{2000}^{\infty} \int_{0}^{y} f_{XY}(x, y)\, dx\, dy \\
&= \int_{2000}^{\infty} \int_{0}^{y} k e^{-0.01x - 0.02y}\, dx\, dy \\
&= k \int_{2000}^{\infty} e^{-0.02y} \left[\int_{0}^{y} e^{-0.01x}\, dx \right] dy \\
&= k \int_{2000}^{\infty} e^{-0.02y} \left[\frac{e^{-0.01x}}{-0.01} \right]_{0}^{y} dy \\
&= k \int_{2000}^{\infty} e^{-0.02y} \cdot 100(1 - e^{-0.01y})\, dy \\
&= 100k \int_{2000}^{\infty} (e^{-0.02y} - e^{-0.03y})\, dy \\
&= 100k \lim_{b \to \infty} \left[\frac{e^{-0.02y}}{-0.02} - \frac{e^{-0.03y}}{-0.03} \right]_{2000}^{b} \\
&= 100k \left[(0 - 0) - \left(\frac{e^{-0.02(2000)}}{-0.02} - \frac{e^{-0.03(2000)}}{-0.03} \right) \right] \\
&= 100k \left(50e^{-40} - \frac{100}{3} e^{-60} \right)
\end{aligned}
$$

Substitute $k = 6 \times 10^{-6}$:

$$
\begin{aligned}
P(Y > 2000) &= 6 \times 10^{-4} \left(50e^{-40} - \frac{100}{3} e^{-60} \right) \\
&= 0.03e^{-40} - 0.02e^{-60}.
\end{aligned}
$$

5.2 Conditional Probability Distributions and Independence

5.2.1 Independence

Definition. Let X and Y be two random variables. Then they are **independent** if and only if

$$f_{XY}(x,\, y) = f_X(x)f_Y(y).$$

Remark. Recall math in high school, we know that

$$P(A \cap B) = P(A)P(B)$$

if both events A and B are independent.

Definition. Let X and Y be two random variables. Then the mean from a joint distribution $(X,\, Y)$ with a transformer function $H(X,\, Y)$ of X and Y is defined as

$$E[H(X,\, Y)] = \sum_{x \in X} \sum_{y \in Y} H(X,\, Y) f_{XY}(x,\, y) \qquad \text{(Discrete case)}$$

$$E[H(X,\, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} H(X,\, Y) f_{XY}(x,\, y)\, dx\, dy \qquad \text{(Continuous case)}$$

Example. Find $\displaystyle\sum_{x \in X} \sum_{y \in Y} x f_{XY}(x,\, y)$.

Sol.

$$\begin{aligned}
\sum_{x \in X} \sum_{y \in Y} x f_{XY}(x,\, y) &= \sum_{x \in X} x \left(\sum_{y \in Y} f_{XY}(x,\, y) \right) \\
&= \sum_{x \in X} x f_X(x) \\
&= E(X).
\end{aligned}$$

Example. Find $\displaystyle\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{XY}(x,\, y)\, dy\, dx$.

Sol.

$$\begin{aligned}
\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{XY}(x,\, y)\, dy\, dx &= \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f_{XY}(x,\, y)\, dy \right) dx \\
&= \int_{-\infty}^{\infty} x f_X(x)\, dx \\
&= E(X).
\end{aligned}$$

5.2.2 Covariance and Correlation

Let's review the variance in one variable.

$$\begin{aligned} Var(X) &= E((X - E(X))^2) \\ &= E(X^2 - 2XE(X) + E^2(X)) \\ &= E(X^2) - 2E^2(X) + E^2(X) \\ &= E(X^2) - E^2(X) \end{aligned}$$

So we have

$$\begin{aligned} Var(X) &= E(X^2) - E^2(X) \\ &= E(X \cdot X) - E(X) \cdot E(X) \\ &\Rightarrow E(X \cdot Y) - E(X) \cdot E(Y). \end{aligned}$$

$E(X \cdot Y) - E(X) \cdot E(Y) = E[(X - E(X))(Y - E(Y))]$, it is covariance(共變異數).

Definition. The **covariance** between the random variables X and Y, denoted as $\text{cov}(X, Y)$ or σ_{XY} is

$$\text{cov}(X, Y) \equiv E[(X - E(X))(Y - E(Y))] = E(XY) - E(X)E(Y).$$

Theorem. Let (X, Y) be a ordered pair random variables with joint probability distribution function $f_{XY}(x, y)$. If X and Y are independent, then $\text{cov}(X, Y) = 0$.

Definition. (Correlation coefficient) The **Pearson correlation coefficient**(皮爾森相關係數) between random variables X and Y, denoted as ρ_{XY}, is

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{Var(X)Var(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

For any two random variables X and Y,

$$-1 \leq \rho_{XY} \leq 1.$$

Theorem. The conditions for **linearity** between X and Y if and only if

$$Y = \beta_0 + \beta_1 X.$$

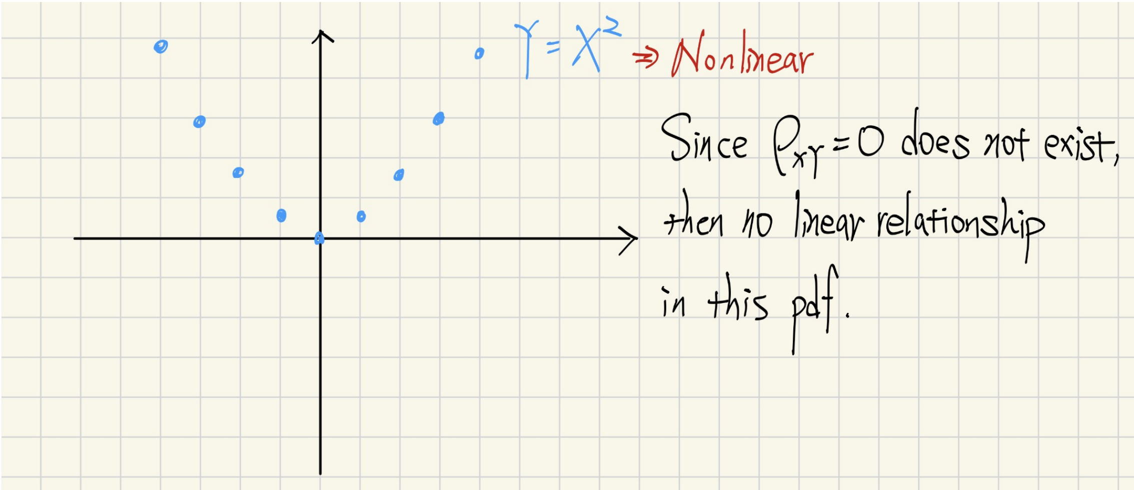
Remark. Let ρ_{XY} be the Pearson correlation coefficient between random variables X and Y.

1. If $\rho_{XY} = 1$, then we call it is perfect positive correlation and $\beta_1 > 0$ for $Y = \beta_0 + \beta_1 X$.

2. If $\rho_{XY} = -1$, then we call it is perfect negative correlation and $\beta_1 < 0$ for $Y = \beta_0 + \beta_1 X$.

3. If $\rho_{XY} = 0$, then we call it is no linear relationship.

Example. Consider $Y = X^2$.



Remark. If X and Y are independent, then $\rho_{XY} = 0$.

5.2.3 Conditional Probability Distributions

Definition. Let (X, Y) be a ordered pair random variables with joint probability distribution function $f_{XY}(x, y)$. Suppose $f_X(x)$ and $f_Y(y)$ are marginal probability distribution function for $f_{XY}(x, y)$.

(1) The conditional density for X given $Y = y$ is defined as

$$f_{X|y}(x) = \frac{f_{XY}(x, y)}{f_Y(y)}.$$

Recall math in high school, we know that

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

(2) The conditional density for Y given $X = x$ is defined as

$$f_{Y|x}(y) = \frac{f_{XY}(x, y)}{f_X(x)}.$$

Example. Suppose

$$f_{XY}(x,\, y) = \frac{c}{x}, \quad c = \frac{1}{6 - 27\ln{(33/27)}},$$

where $27 \le y \le x \le 33$. Find

(a) $f_X(x)$

(b) $f_Y(y)$

(c) $f_{X|y}(x)$

(d) The probability that $x > 32$ given $y = 30$.

(e) The expected value of X given $y = 30$.

Sol. We don't substitute c into the function since it is too difficult to evaluate.

(a)

$$\begin{aligned} f_X(x) &= \int_{27}^{x} f_{XY}(x,\, y)\, dy \\ &= \int_{27}^{x} \frac{c}{x}\, dy \\ &= \frac{cy}{x}\Big|_{27}^{x} \\ &= c\frac{x - 27}{x}, \text{ where } 27 \le x \le 33. \end{aligned}$$

(b)

$$\begin{aligned} f_Y(y) &= \int_{y}^{33} f_{XY}(x,\, y)\, dx \\ &= \int_{y}^{33} \frac{c}{x}\, dx \\ &= c(\ln 33 - \ln y) \\ &= c\ln\frac{33}{y}, \text{ where } 27 \le y \le 33. \end{aligned}$$

(c)

$$\begin{aligned} f_{X|y}(x) &= \frac{f_{XY}(x,\, y)}{f_Y(y)} \\ &= \frac{c/x}{c(\ln 33 - \ln y)} \\ &= \frac{1}{x(\ln 33 - \ln y)} \\ &= \frac{1}{x\ln(33/y)} \end{aligned}$$

(d)

$$\begin{aligned}
P(X > 32|y = 30) &= \int_{32}^{33} \frac{1}{x(\ln 33/30)}\,dx \\
&= \left[\frac{\ln|x|}{\ln(33/30)}\right]_{32}^{33} \\
&= \frac{\ln 33 - \ln 32}{\ln(33/30)} \\
&= \frac{\ln(33/32)}{\ln(33/30)}
\end{aligned}$$

(e)

$$\begin{aligned}
E(X|y = 30) &= \int_{27}^{33} x f_{X|y=30}(x)\,dx \\
&= \int_{27}^{33} x \frac{1}{x\ln(33/30)}\,dx \\
&= \int_{27}^{33} \frac{1}{\ln(33/30)}\,dx \\
&= \frac{6}{\ln(33/30)}.
\end{aligned}$$